NCAA Basketball Tournament Prediction Modeling

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1,449	961	4
Men's Games	Women's Games	Models Tested

Project Overview

PROJECT GOAL

- Predict NCAA tournament game outcomes for both men's and women's brackets.
- Compare multiple classification methods to find the best predictive model.
- Use cross-validation to ensure selection is based on out-of-sample performance.
- Test at least one logistic regression and one tree-based model as required.

8-PHASE WORKFLOW

1 Data Inspection	2 Feature Engineering
3 Exploratory Analysis	4 Baseline Models
5 Expanded Comparison	6 Cross-Validation
7 Final Model Selection	**8 Conclusions**

Dataset Background

Men's Tournament	
Rows	**1,449**
Columns	**40**
Win Balance	**A: 727 | B: 722**

Women's Tournament	
Rows	**961**
Columns	**40**
Win Balance	**A: 486 | B: 475**

Key Design Detail

Teams were randomly assigned to Team A and Team B in each game so the winner was not always listed first. This required a difference-based predictor approach rather than using raw team statistics directly.

Feature Engineering

APPROACH

Response Variable

y = 1 if Team A wins, y = 0 if Team B wins

Predictor Creation

Each predictor was computed as the difference between Team A's average statistic minus Team B's average statistic. A positive value means Team A had the advantage.

Cleaned datasets renamed: men2 and women2

15 DIFFERENCE PREDICTORS

SeedDiff	FGMDiff	FGADiff
ScoreDiff	FGM3Diff	FGA3Diff
AstDiff	FTMDiff	FTADiff
TODiff	ORDiff	DRDiff
StlDiff	BlkDiff	PFDiff

Exploratory Data Analysis

Class Balance

Win rates were nearly 50/50 in both datasets. Men: 727 A-wins vs 722 B-wins. Women: 486 vs 475. No class imbalance correction was needed.

Histograms

SeedDiff and ScoreDiff were centered near zero in both datasets. The women's ScoreDiff showed a slightly wider spread than the men's version.

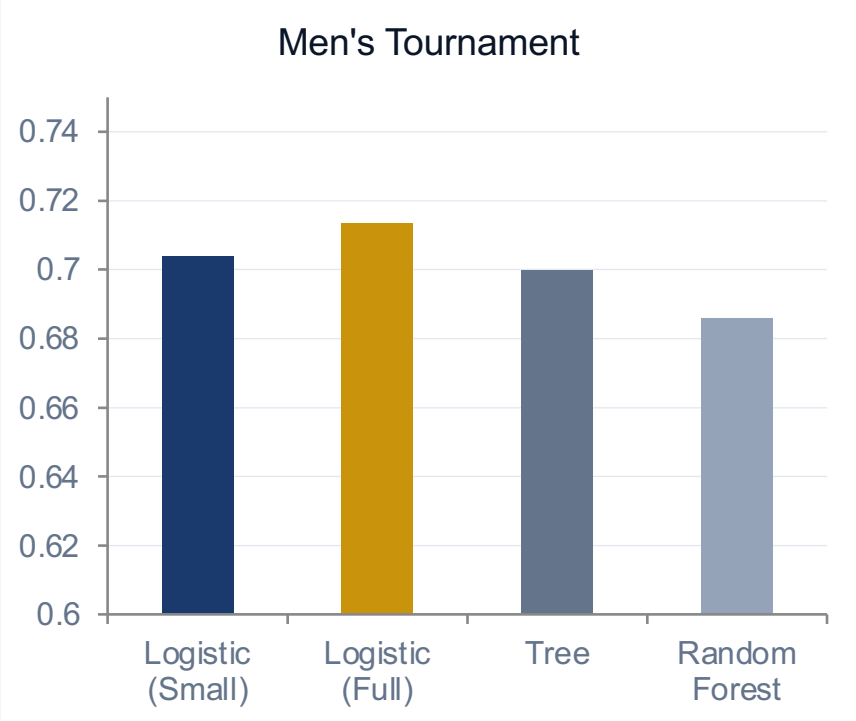
Boxplots

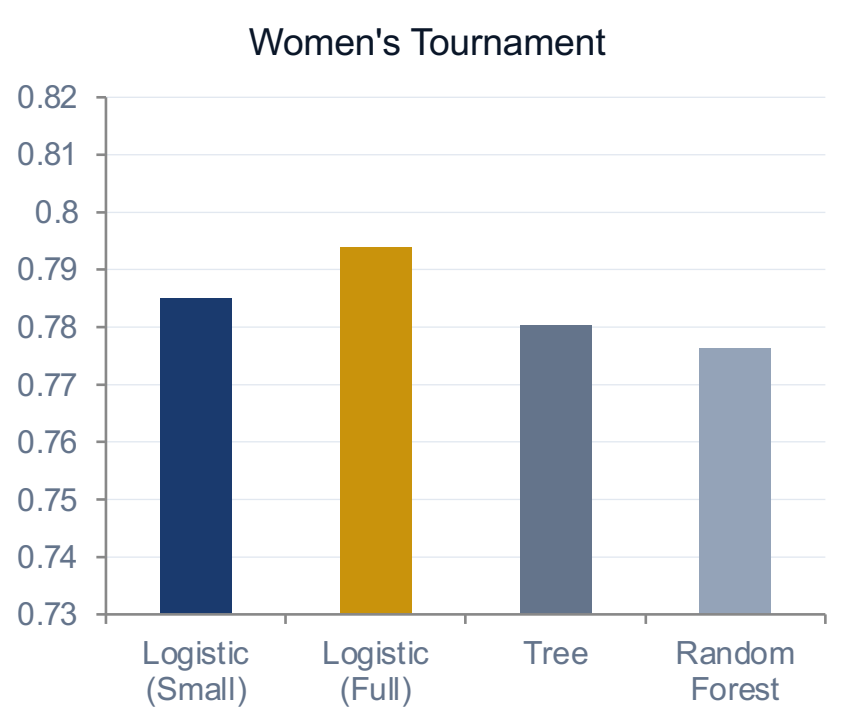
Higher ScoreDiff and better SeedDiff were both associated with Team A wins. These patterns aligned with basketball intuition about team quality.

Correlations

Moderate correlations were found among assists, rebounds, and shooting-related variables. This motivated comparing a small versus full logistic regression.

Model Comparison: Training Accuracy





10-Fold Cross-Validation: ROC AUC

Model	Men ROC	Women ROC	Best Model?
Logistic (Small)	0.7771	0.8714	
Logistic (Full)	**0.7820**	**0.8693**	**BEST**
Tree	0.7117	0.8291	
Random Forest	0.7522	0.8607	

Final Model Selection

Full Logistic Regression chosen for both men's and women's tournaments

01

Best Cross-Validated ROC

Outperformed all other models using proper out-of-sample evaluation via 10-fold cross-validation.

02

Consistent Across Both Datasets

Achieved the top ROC for men (0.7820) and women (0.8749) with no competing method finishing higher in either.

03

Easier to Interpret

Logistic regression coefficients are interpretable, making results easier to explain compared to random forest.

04

More Flexible is Not Always Better

Random forests and trees did not outperform logistic regression, confirming that model complexity alone is not sufficient.

Applying the Model: 2026 Predictions

1

List matchups

Identify all 2026 bracket pairings and label Team A and Team B.

2

Compute differences

Calculate the same 15 SeedDiff, ScoreDiff, and stat difference predictors used in training.

3

Apply model

Plug the computed predictors into the final full logistic regression model.

4

Get probabilities

Generate a win probability for Team A in each game and use that to guide picks.

Sample Output (Clemson as Team A) **Men: 60.1% win probability** **Women: 77.4% win probability**

Conclusions & Limitations

KEY TAKEAWAYS

- Difference-based predictors were the most important modeling decision.
- Simpler logistic regression outperformed more complex tree methods.
- Cross-validation was essential for reliable model selection.
- Women's models consistently outperformed men's across all methods.
- Both separate models are now ready for 2026 bracket predictions.

LIMITATIONS

No Injury Data

Player availability and injuries are not captured in the season-average statistics.

No Momentum Signal

Late-season performance trends and coaching changes are not reflected in the model.

Tournament Variance

March Madness upsets are inherently unpredictable and no model eliminates that uncertainty.